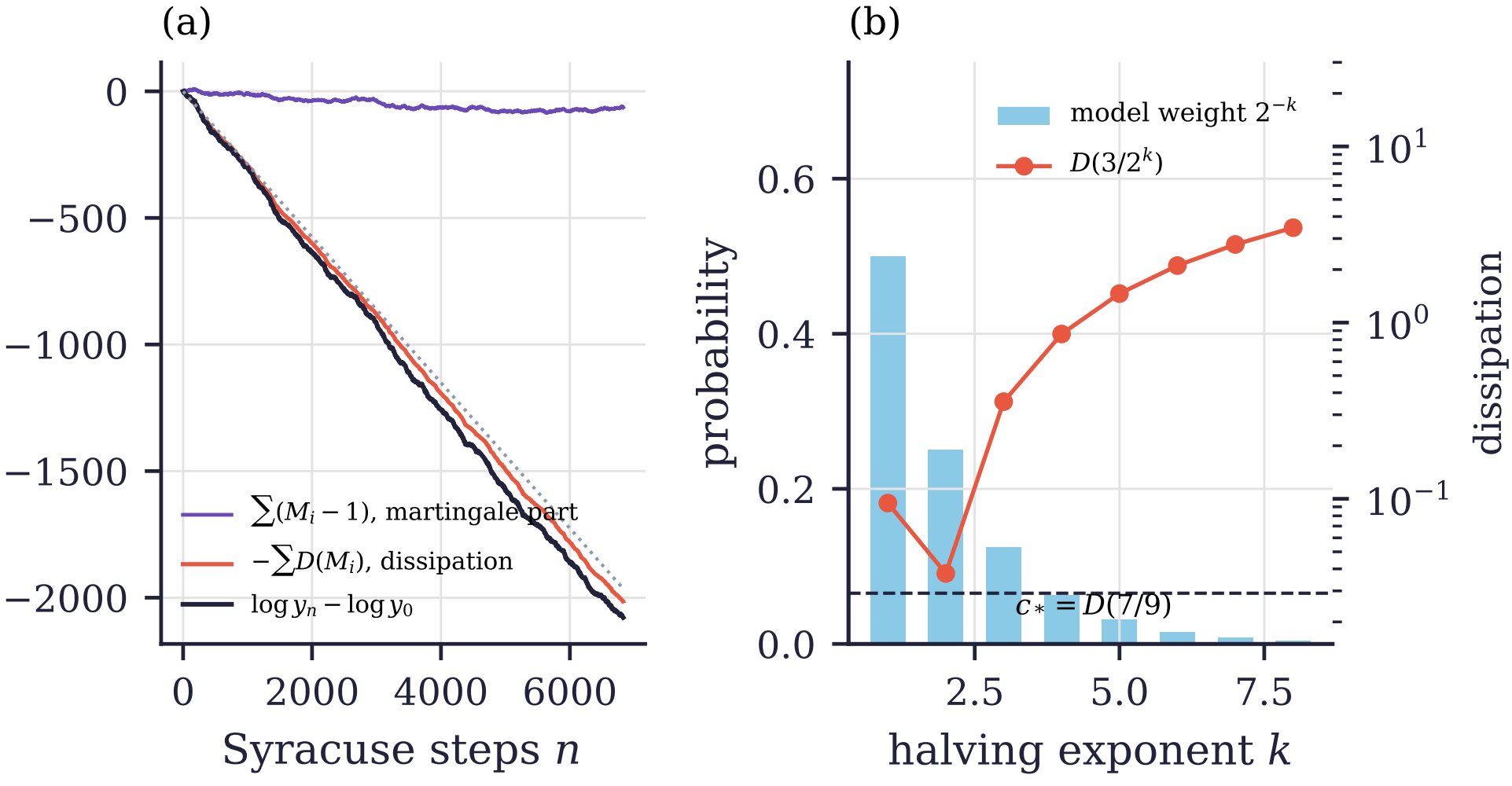
(a)
0
−500
−1000
−1500
−2000
0
2000
4000
6000
Syracuse steps n
$\sum(M_i - 1)$, martingale part
$-\sum D(M_i)$, dissipation
$\log y_n - \log y_0$
(b)
probability
0.0
0.2
0.4
0.6
2.5
5.0
7.5
halving exponent k
model weight 2^{-k}
$D(3/2^k)$
$c_* = D(7/9)$
dissipation
10^1
10^0
10^{-1}